

PIROVSKOYE, Russian Federation

WMO: 293630

Lat: 57.6312N

Lon: 92.2714E

Elev: 179

StdP: 99.19

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-38.5	-44.3	0.0	-41.6	-40.5	0.1	-37.7	7.4	-8.6	6.7	-9.7	0.9	60	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	29.1	18.9	27.2	18.2	25.4	17.5	20.3	26.0	19.5	25.0	18.6	23.7	2.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.7	22.6	17.5	12.8	21.7	16.6	12.1	21.0	59.6	26.0	56.7	25.0	53.5	23.8	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.9	6.1	5.3	-46.1	31.1	2.5	1.6	-47.9	32.3	-49.3	33.2	-50.7	34.1	-52.5	35.3		
			-44.9	21.8	1.6	1.2	-46.1	22.7	-47.0	23.4	-47.9	24.1	-49.0	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-21.0	-15.3	-7.2	1.6	8.3	15.9	17.6	15.1	7.5	-0.6	-12.6	-16.3
	DBStd	14.82	10.15	8.17	7.43	4.85	5.49	4.50	3.47	3.74	4.28	5.76	9.48	9.99
	HDD10.0	4493	960	709	532	254	103	9	0	5	98	331	678	816
	HDD18.3	6944	1218	942	790	502	316	96	56	112	324	588	928	1074
	CDD10.0	662	0	0	0	2	48	186	236	164	24	1	0	0
	CDD18.3	72	0	0	0	0	3	23	33	13	0	0	0	0
	CDH23.3	879	0	0	0	1	41	347	348	140	3	0	0	0
	CDH26.7	202	0	0	0	0	7	86	83	26	0	0	0	0
Wind	WSAvg	2.4	2.3	2.4	2.6	3.0	2.9	2.2	1.9	1.9	2.2	2.5	2.7	2.6
Precipitation	PrecAvg	628	37	29	30	40	51	63	62	68	65	66	65	52
	PrecMax	793	78	48	59	67	102	111	124	106	114	96	107	87
	PrecMin	474	2	13	6	16	20	7	15	32	15	20	26	15
	PrecStd	82	14	9	12	13	20	23	26	20	22	17	17	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	1.2	9.0	19.1	28.0	31.4	30.7	29.3	23.3	16.0	6.9	1.0
		MCWB	-1.8	-1.4	3.5	10.3	15.0	18.8	20.6	20.2	14.8	8.8	5.1	-0.1
	2%	DB	-2.7	-0.6	6.0	15.4	24.0	29.2	29.2	26.8	19.7	11.7	3.8	-0.8
		MCWB	-3.8	-2.3	1.7	8.0	14.2	18.2	19.6	19.1	13.6	7.0	2.0	-1.5
	5%	DB	-5.0	-2.0	4.3	12.2	21.1	27.1	27.2	24.6	17.3	9.0	2.0	-2.3
		MCWB	-5.7	-3.3	0.8	6.1	12.4	17.3	19.0	18.0	12.4	6.0	0.6	-2.9
	10%	DB	-7.3	-3.7	2.5	9.7	18.4	25.2	25.2	22.6	14.8	6.7	-0.1	-4.1
		MCWB	-8.0	-4.8	-0.2	4.8	11.2	16.8	18.1	17.1	11.0	4.7	-0.9	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-0.7	4.5	10.8	16.8	20.6	22.7	21.6	15.7	9.6	5.6	0.2
		MCDB	-0.4	0.5	8.3	16.5	25.8	27.6	29.6	26.2	20.3	14.5	7.4	0.7
	2%	WB	-3.8	-2.1	2.4	8.5	14.8	19.5	20.7	19.9	14.4	7.6	2.4	-1.4
		MCDB	-2.9	-0.9	5.2	14.1	22.0	26.1	26.3	25.2	18.5	10.9	3.6	-0.9
	5%	WB	-5.7	-3.2	1.1	6.7	13.2	18.7	19.9	18.9	13.1	6.2	0.7	-3.0
		MCDB	-4.9	-1.8	3.5	11.4	19.9	25.0	25.2	23.7	17.1	8.7	1.8	-2.4
	10%	WB	-8.0	-4.9	0.0	5.1	11.7	17.7	18.9	17.7	11.4	5.0	-0.9	-4.7
		MCDB	-7.2	-3.6	2.5	9.3	17.8	23.5	23.7	21.6	14.2	7.1	0.0	-4.1
Mean Daily Temperature Range	5% DB	MDBR	8.4	10.1	10.9	9.5	10.8	11.9	11.1	10.5	8.9	6.9	7.1	7.7
		MCDBR	8.0	8.5	11.1	13.8	15.4	16.2	14.6	13.9	14.0	10.5	6.2	7.5
		MCWBR	7.5	7.4	8.0	8.0	7.7	6.6	6.4	6.7	8.0	6.9	5.3	7.0
	5% WB	MCDBR	7.8	7.9	9.3	12.5	13.7	13.0	12.0	11.9	12.4	9.4	6.5	7.2
		MCWBR	7.4	7.1	7.0	7.7	7.4	5.9	5.9	6.1	7.5	6.7	5.8	6.9
Clear-Sky Solar Irradiance	taub	0.284	0.317	0.356	0.386	0.403	0.412	0.434	0.408	0.365	0.342	0.306	0.278	
	taud	1.974	1.972	1.944	2.068	2.221	2.281	2.227	2.278	2.361	2.244	2.021	1.930	
	Ebn at Noon	519	679	762	814	830	827	796	791	773	665	484	394	
	Edh at Noon	62	98	133	138	127	122	126	112	89	75	59	48	
All-Sky Solar Radiation	RadAvg	0.66	1.60	2.95	4.42	5.00	5.97	5.48	4.24	2.55	1.25	0.61	0.37	
	RadStd	0.09	0.13	0.22	0.29	0.52	0.56	0.38	0.35	0.35	0.12	0.11	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page